Alvaro Perdomo Gutierrez

Software Development Engineer in Test

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PROFESSIONAL SUMMARY

Senior Software Engineer with **12+** years across fintech, healthcare, and gaming, building robust automated testing frameworks and scalable QA infrastructure. Experienced in test automation architecture, CI/CD integration, and API testing using Playwright, JavaScript, and modern development practices. Proven ability to lead testing initiatives in Agile teams and deliver reliable, maintainable automation solutions.

SKILLS

Languages: JavaScript, TypeScript, Java, Python, PHP, SQL, HTML, CSS
Testing: Playwright, Cypress, Selenium, Cucumber, Page Object Model, Jest, Pytest, TestNG, Mocha
Backend: Node.js, Express, FastAPI, Django, Laravel, Spring Boot
Frontend: React, Vue.js, Next.js, Angular
Database: PostgreSQL, MySQL, MongoDB, Redis, TcaplusDB
Cloud & DevOps: GitHub Actions, Jenkins, CircleCI, Docker, Kubernetes, AWS, Google Cloud, Terraform
Others: RESTful API, GraphQL, gRPC, WebSocket, Grafana, Prometheus, Datadog, Git, Jira, Confluence, Agile, Scrum

EXPERIENCE

Leanware May 2024–Present
Senior Software Engineer

- Led the design and implementation of an end-to-end test automation framework for an AI-powered dental evaluation platform, leveraging **Playwright** and **JavaScript** to validate complex computer vision workflows and diagnostic accuracy across 40+ test cases.
- Championed the integration of automated testing into **GitHub Actions** CI/CD pipelines, reducing manual regression testing cycles from 6 hours to 45 minutes and enabling daily deployments for three concurrent fintech and healthcare projects.
- Built a comprehensive Page Object Model architecture in **Playwright** that abstracted cloud infrastructure dependencies and API contracts, making test suites maintainable across 15 team members and reducing test maintenance overhead by 30%.
- Implemented API testing workflows in **JavaScript** and **TypeScript** to validate bidirectional integrations between cloud services and databases, catching schema misalignment issues before they reached staging environments.
- Collaborated with backend engineers to align test contracts with **REST API** specifications, ensuring automated test assertions matched actual service behavior and eliminating false-positive test failures.

- Designed accessibility and security test scenarios for a financial platform, incorporating automated checks for WCAG compliance and SQL injection patterns that improved audit readiness and reduced security-related production incidents by 40%.
- Mentored two junior QA engineers on test automation fundamentals and **Playwright** best practices, establishing peer code review standards that raised test code quality and coverage across the team.
- Documented test architecture and automation runbooks in **Confluence**, enabling on-call engineers to troubleshoot and extend test suites independently during incident response.
- Participated in architecture discussions for infrastructure testing, advocating for containerized test environments in **Docker** that increased test parallelization and reduced CI/CD runtime variance by 25%.
- Resolved critical flaky test issues by refactoring assertion logic and adding explicit waits, stabilizing test pass rates from 82% to 97% and restoring team confidence in automated test signals.

ThredUp Mar 2022–May 2024

Senior Software Engineer

- Spearheaded the development of automated test coverage for a machine learning-powered image recognition and visual search platform, building **Playwright** test suites that validated image processing pipelines handling 100K+ photos daily.
- Designed and built an API test automation framework in **JavaScript** and **Node.js** to validate ML model inference endpoints and image tagging workflows, ensuring consistent visual recognition quality across e-commerce catalog integrations.
- Integrated automated testing into **GitHub Actions** pipelines for the Databricks platform migration project, establishing test gates that prevented model deployment regressions and reduced time-to-ML-production from 2 quarters to 1 month.
- Implemented **Cypress** end-to-end tests for the AI Shopping and Style Chat features launched in the mobile app, validating complex user workflows involving image search, personalization, and real-time recommendations.
- Built monitoring and alerting dashboards in **Prometheus** and **Grafana** to track test automation health metrics and CI/CD execution trends, surfacing test execution bottlenecks that improved pipeline throughput by 20%.
- Collaborated with data science teams to design test strategies for auto-photography and resale-pricing ML models, creating synthetic datasets and automation assertions that validated model behavior across 172M+ inventory items.
- Reduced test maintenance burden by introducing a **Cucumber** BDD framework and shared step definitions that non-technical stakeholders could read and extend, increasing test transparency across product and engineering teams.
- Debugged and resolved flaky tests in high-volume transaction scenarios by analyzing test logs and adding strategic waits, achieving 96% stable pass rates in the seller processing automation suites.

Spring Health

Nov 2020–Mar 2022

Software Engineer

- Played a key role in building automated test frameworks for a precision-matching ML platform supporting 2M+ members across Fortune 500 accounts, using **Playwright** and **JavaScript** to validate clinical data workflows and member eligibility rules.
- Designed and executed API test automation strategies for the Care Navigation platform during its global expansion, writing **Selenium** tests in **Java** that verified multi-language support, jurisdiction-aware logic, and dependent eligibility matching across 12+ regions.
- Implemented comprehensive test coverage for HIPAA-compliant infrastructure scaling projects, ensuring automated tests validated encryption standards, audit logging, and data isolation requirements that supported PepsiCo, Instacart, and Bain onboarding.
- Built test utilities in **Python** and **FastAPI** to mock clinical data pipelines and ML model predictions, enabling fast feedback loops for operational ML features like dropout-risk prediction and demand forecasting.
- Collaborated with platform engineers to establish CI/CD best practices using **Jenkins** and **Docker**, standardizing test environment provisioning and reducing test execution variance in production-like staging setups.
- Participated in load and performance testing initiatives for the operationalized ML platform, using **k6** and **Grafana** to identify bottlenecks in the precision-matching algorithm that processed millions of clinical data points daily.

Tencent Games

Nov 2014–Nov 2020

Software Engineer

- Built game testing infrastructure and automated test suites in **Java** for Honor of Kings, a mobile MOBA that scaled from launch to 100M DAU, validating gameplay mechanics, matchmaking systems, and live service updates.
- Designed and implemented anti-cheat test automation using kernel-level monitoring and **C++**, ensuring automated tests could verify ACE anti-cheat effectiveness against known exploit patterns and maintain competitive integrity.
- Contributed to test strategies for PUBG Mobile and Arena of Valor, writing **Selenium** and custom C++ test utilities that validated cross-platform multiplayer synchronization and real-time gameplay state consistency across millions of concurrent players.
- Implemented API testing workflows for Tencent Gaming Buddy emulator, building test harnesses that validated emulator compatibility with UE4 game engines and TcaplusDB backend services handling 40M+ daily concurrent users.
- Collaborated with game operations teams to establish automated regression testing for live service patches, reducing post-deployment incident detection time from 4 hours to 15 minutes and improving player experience stability.
- Participated in infrastructure testing for GCloud platform integrations and TcaplusDB clusters, writing tests that verified database failover behavior and connection pooling under high-concurrency load scenarios typical of live multiplayer gaming.

EDUCATION

The Hong Kong Polytechnic University Sep 2010–Nov 2014

Bachelor of Science in Computer Science